Joshua N. Grant

Geospatial Systems Architect | Senior Data Engineer | Cloud Data Platform Builder

Knoxville, TN | jngrant@live.com | +1 (865) 803-3495 | github.com/sempervent | linkedin.com/in/joshuanagrant | notjustadatum.blogspot.com

---

Professional Summary

Geospatial systems architect and senior data engineer with experience designing production data platforms, cloud-native geospatial pipelines, spatial decision support tools, and research-to-production software systems. Strong background in Python, SQL, AWS, Airflow, Prefect, PostGIS, GeoPandas, raster/vector analytics, containerized deployment, and cross-functional technical leadership. Known for converting experimental workflows, scientific models, and legacy data processes into maintainable applications, automated pipelines, and scalable data products.

Technical Skills

Languages: Python, SQL, R, JavaScript, Bash, PHP, LaTeX

Cloud & Infrastructure: AWS S3, Lambda, EC2, RDS, Fargate; GCP; Azure; Docker; Kubernetes; Ansible; GitHub Actions; AWS CodePipeline

Data Engineering: Airflow, Prefect, Kafka, ETL/ELT pipelines, data modeling, workflow orchestration, message queues, Spark, Slurm, Hadoop, NiFi

Geospatial Systems: PostGIS, GeoPandas, GeoParquet, GDAL, Rasterio, Pyproj, QGIS, ArcGIS, Mapbox, Leaflet, Martin, zonal statistics, raster/vector processing

Databases: PostgreSQL, TimescaleDB, MongoDB, MySQL, Snowflake, SQLite, SpatiaLite, DuckDB, Redis, RabbitMQ

Applications & APIs: FastAPI, Flask, Django, REST API development, Swagger/OpenAPI, Shiny, Electron

Analytics & Visualization: Scikit-learn, TensorFlow, Pandas, NumPy, matplotlib, Plotly, Grafana, Kibana, Leaflet, model deployment

Leadership: Technical architecture, team leadership, software quality assurance, stakeholder collaboration, Agile delivery, documentation

Professional Experience

Geospatial Systems Architect 2023–Present
Oak Ridge National Laboratory Oak Ridge, TN

- Architect and maintain GeoParquet-based geospatial data warehouse patterns for risk component storage, retrieval, analysis, and downstream application delivery.
- Design and support a geospatial decision support tool deployed across web and desktop environments, aligning shared data models, APIs, and user-facing workflows.
- Build cloud-oriented raster ingestion and analysis pipelines using Prefect, Python, object storage, and geospatial processing libraries.
- Develop high-sensitivity parcel tracking and monitoring applications using IoT data streams, Kafka, TimescaleDB, Airflow, and AWS-hosted services.
- Design geospatial databases and processing workflows for large raster collections, vector geometry aggregation, and zonal statistics at operational scale.
- Modernize legacy Python systems by replacing brittle custom patterns with standard libraries, clearer interfaces, repeatable workflows, and maintainable project structure.
- Lead technical planning and implementation across backend, frontend, database, and pipeline components for geospatial and time-series data products.

Senior Software Engineer – Data Engineering 2022–2023
Bold Penguin Columbus, OH / Remote

- Designed centralized development and deployment workflows using GitHub Actions, AWS services, reusable Python modules, and automated release practices.
- Led migration of deployments, flows, and tasks from Prefect 1 to Prefect 2, improving maintainability and aligning orchestration patterns across teams.
- Built Prefect task execution patterns that routed workloads to EC2 or Fargate based on runtime conditions such as file size and processing requirements.
- Created EC2 and Fargate-backed microservices for document conversion, data extraction, and backend processing workflows.
- Designed and maintained REST APIs that supported stakeholder requirements, client integrations, and production data exchange.
- Implemented scalable message queue architecture for high-load data extraction services, improving throughput and resilience of processing operations.
- Configured GitHub Actions and AWS CodePipeline to automate build, test, and deployment workflows for Python and JavaScript services.

Data Engineer 2019–2022
Oak Ridge National Laboratory Oak Ridge, TN

- Replaced a brittle cron-based web scraping process with a more reliable Airflow ETL workflow deployed in Kubernetes and cloud infrastructure.
- Developed and maintained full-stack Docker images using Python, R, Airflow, and Shiny to automate spatial-temporal data acquisition for WSTAMP and Hadoop data warehouse storage.
- Led development for a COVID-19 tracking project using R, Python, downstream model creation, geospatial analysis, and operational reporting workflows.
- Built containerized Airflow ETL workflows, Python microservices, and Swagger/OpenAPI REST APIs for data exchange with PostgreSQL, AWS S3, AWS RDS, and MongoDB.
- Automated web scraping and social media ingestion for misinformation classification, statistical modeling, network analysis, Elasticsearch indexing, and Kibana visualization.
- Served as Software Quality Assurance board chair for the Geospatial Science and Human Security Division.
- Contributed to ORNL Data Council discussions on lab-wide data management, architecture, governance, and sharing practices.

Data Scientist II Subcontractor 2018–2019
Oak Ridge National Laboratory Oak Ridge, TN

- Automated WSTAMP web application data ingestion using R, Python, Airflow, and Docker.
- Authored an R package to normalize geographic identifiers and unify data access across online APIs.
- Developed a SpatiaLite-backed SQLite database to simplify lookup, retrieval, and integration of geographic data.
- Built a Python zonal statistics application for calculating raster summaries across shapefile geometries.

Owner & Contractor-For-Hire 2018
Specrabella Knoxville, TN

- Designed, developed, and deployed an interactive Leaflet-based energy informatics application integrated with Snowflake and machine-learning-derived calculations.
- Consulted with clients on ground-up web application architecture, database design, data visualization, and deployment needs.
- Built data ingestion workflows using cURL, PHP, R, Python, and Snowflake to scrape, clean, store, and visualize energy datasets.
- Streamlined training and development workflows with Ansible, Vagrant, PHP, and Bash.
- Designed PostgreSQL databases for energy and financial data products.

Bioinformatician, Project Manager, & Technical Sales Representative 2016–2018
Microbial Insights Knoxville, TN

- Established an automated NGS analysis pipeline using R, MySQL, Python, Bash, JavaScript, and Shiny-backed internal tooling.
- Automated client-facing statistical workflows including linear models, ANOVA, self-organizing maps, clustering, PCA, and PCoA.
- Developed an ETL customer data visualization tool using R, PHP, jQuery, and MySQL for dynamic qPCR result analysis.
- Built a custom R package and local Shiny applications to generate client-ready LaTeX PDF reports more efficiently.
- Constructed and maintained a PostgreSQL- and PHP-backed intranet wiki supporting project management and customer service operations.

Graduate Research Assistant 2014–2016
Dr. Neal Stewart's Plant Biotechnology Lab, University of Tennessee Knoxville, TN

- Analyzed cell suspension experiments using linear models, ANOVA, PCA, Python, and R for DOE reporting and scientific communication.
- Evaluated suspension cultures through chemical and spectral processes for lignin formation and summarized findings for research deliverables.
- Collaborated on a novel single-cell suspension and cryopreservation robotic system.
- Executed monocot genetic modification workflows for *Panicum*, *Oryza*, *Sorghum*, *Saccharum*, and *Zea mays*.

Laboratory Assistant 2012–2014
Dr. Neal Stewart's Plant Biotechnology Lab, University of Tennessee Knoxville, TN

- Developed Python-based statistical methodology for screening lignin content and imaging cell characteristics both *in vivo* and *in vitro*.
- Extracted genomes from NCBI, cleaned and normalized datasets, and performed exploratory data analysis using Python.
- Supported USDA-APHIS-compliant research involving transgenic plants.
- Implemented an *E. coli* bioreactor for protein production workflows.

Laboratory Assistant 2002–2012
Dr. Paris Lambdin's Biosystematics and Biological Control Lab, University of Tennessee Knoxville, TN

- Designed instructional modules for undergraduate and graduate courses using HTML, CSS, JavaScript, and Flash.
- Collected field samples and research data for biological control and biosystematics projects.
- Supported community outreach programs including Bloomsdays, Buggy Buffet, and 4-H Camps.
- Performed forest coverage analysis using ArcGIS.

Selected Projects

Final Fantasy Football — Fantasy football semantic learning and predictive analytics application.
Technologies: Python, Scikit-learn, Pandas, NumPy, Flask, Docker, Airflow

Where I've Been — Travel history visualization application with spatial data storage and web mapping.
Technologies: Python, Flask, Docker, PostgreSQL, Leaflet, Vue, Bootstrap

Decentralized Content Reward System — Prototype web platform for decentralized content reward workflows.
Technologies: Python, Rust, Flask, Docker, PostgreSQL, React, Bootstrap

This Is A Casino — Experimental trading and market visualization application using machine learning and reinforcement learning workflows.
Technologies: Python, Flask, Docker, PostgreSQL, React, Bootstrap

---

Education

Master of Science in Plant Sciences – Plant Molecular Genetics
University of Tennessee — Spring 2017
GPA: 3.72/4.0

Bachelor of Science in Plant Sciences – Biotechnology
University of Tennessee — Spring 2014
GPA: 3.74/4.0; Magna Cum Laude

Publications

1. Peluso, A., Rastogi, D., Klasky, H. B., Logan, J., Maguire, D., Grant, J., Christian, B., & Hanson, H. A. Environmental determinants of health: Measuring multiple physical environmental exposures at the United States census tract level. *Health & Place*, 89, 103303, 2024.
2. Brelsford, C., Tennille, S., Myers, A., Chinthavali, S., Tansakul, V., Denman, M., Coletti, M., Grant, J., Lee, S., Allen, K., et al. A dataset of recorded electricity outages by United States county 2014–2022. *Scientific Data*, 11, 271, 2024.
3. Tansakul, V., Myers, A., Tennille, S., Denman, M., Hamaker, A., Huihui, J., Medlen, K., Allen, K., Redmon, D., Chinthavali, S., et al. EAGLE-I Power Outage Data 2014–2022. Oak Ridge National Laboratory technical report, 2023.
4. Stewart, R., Erwin, S., Piburn, J., Nagle, N., Kaufman, J., Peluso, A., Christian, J. B., Grant, J., Sorokine, A., & Bhaduri, B. Near real time monitoring and forecasting for COVID-19 situational awareness. *Applied Geography*, 146, 102759, 2022.
5. Herrmannova, D., Thakur, G., Grant, J., Tansakul, V., Eaton, B., Kotevska, O., Burdette, J., Smyth, M., & Smith, M. Challenges in automated detection of COVID-19 misinformation. OSTI technical report, 2021.
6. Willis, J. D., Grant, J. N., Mazarei, M., Kline, L. M., Rempe, C. S., Collins, A. G., Turner, G. B., Decker, S. R., Sykes, R. W., Davis, M. F., et al. The TcEG1 beetle cellulase produced in transgenic switchgrass is active at alkaline pH and auto-hydrolyzes biomass for increased cellobiose release. *Biotechnology for Biofuels*, 10, 230, 2017.
7. Grant, J. N., Burris, J. N., Stewart, C. N., & Lenaghan, S. C. Improved tissue culture conditions for the emerging C4 model *Panicum hallii*. *BMC Biotechnology*, 17, 39, 2017.
8. Grant, J. N. Evaluation of Hall's Panicgrass (*Panicum hallii* Vasey) as a model system for genetic modification of recalcitrance in switchgrass (*Panicum virgatum* (L.)). Master's thesis, University of Tennessee, Knoxville, 2017.
9. Liu, J., Chen, N., Grant, J. N., Cheng, Z.-M., Stewart, C. N., Jr., & Hewezi, T. Soybean kinome: Functional classification and gene expression patterns. *Journal of Experimental Botany*, 66(7), 1919–1934, 2015.